

Deep Learning Approaches for Sleep Apnea Classification from Polysomnographic EEG Signals

Shashank Manjunath and Mukesh Cheemakurthi

Khoury College of Computer Sciences

Northeastern University

Boston, USA

{manjunath.sh, cheemakurthi.m}@northeastern.edu

Aarti Sathyanarayana

Khoury College of Computer Sciences

Bouvé College of Health Sciences

Northeastern University

Boston, USA

a.sathyanarayana@northeastern.edu

Abstract—Sleep apnea diagnosis via polysomnography remains resource intensive and relies on time consuming manual data analysis and scoring. Recent work has demonstrated that central nervous system effects of sleep apnea events can be detected through electroencephalogram (EEG) signals. However, most work uses a single feature type on various datasets combined with different classification algorithms. In this work, we present a comprehensive comparison of deep learning architectures and feature representations for automated sleep apnea detection from multichannel EEG on a single dataset of pediatric subjects. We evaluate Vision Transformers and Graph Attention Networks across distinct signal representations: raw temporal signals, short-time Fourier transform spectrograms, coherence based graphs, and two topological data analysis (TDA) derived features. Using age and sex matching of our train and test sets, we train on 2410 pediatric subjects and test on 575 pediatric subjects. We achieve a best test AUC of 0.750 using a vision transformer based model trained on TDA features. Stratified analysis across patient demographics (age, sex, AHI severity) and sleep stages (N1, N2, N3, REM) reveals significant performance variation. Our results demonstrate the feasibility of EEG based automated OSA screening while highlighting essential challenges for clinical deployment.

I. INTRODUCTION

Obstructive sleep apnea (OSA) is a highly prevalent sleep disorder affecting an estimated 936 million adults and up to 6.4% children worldwide [2], [20]. Sleep apnea is characterized by repeated episodes of complete or partial upper airway obstruction during sleep. OSA leads to intermittent hypoxia, sleep fragmentation, and excessive daytime sleepiness. Left untreated, OSA significantly increases the risk of cardiovascular disease and all-cause mortality [18].

Despite its clinical significance, OSA remains underdiagnosed [12]. Current diagnosis relies on overnight polysomnography (PSG), which monitors multiple physiological signals including electroencephalogram (EEG), electrocardiogram (ECG), respiratory effort, and oxygen saturation. While PSG studies are the gold standard for OSA classification, these studies require expensive equipment, dedicated sleep laboratories, and trained technicians to manually score sleep stages and respiratory events [26]. This process is time consuming, labor intensive, and subject to interrater variability [17]. These limitations create significant bottlenecks in diagnostic workflows and restrict access to timely OSA diagnosis, particularly in resource-limited settings.

While OSA is fundamentally a respiratory disorder, the symptoms and effects of sleep apnea manifest through central nervous system (CNS) dysfunction. Recurrent apnea events trigger brief awakenings that fragment sleep architecture and prevent restorative deep sleep and REM sleep. These repeated arousals, coupled with intermittent hypoxia, lead to chronic sleep disruption that manifests as excessive daytime sleepiness, impaired cognitive function, mood disturbances, and reduced quality of life [32]. Moreover, emerging evidence suggests that chronic OSA-induced hypoxia may cause long-term neurological damage, including gray matter loss in regions critical for memory and executive function [19].

Electroencephalograms (EEGs) directly measure these CNS effects through electrical activity of the brain, providing unique insights into the effect of apnea events on the brain. Unlike peripheral sensors that detect respiratory obstruction (nasal airflow, thoracic effort) or its immediate physiological consequences (oxygen desaturation), EEG captures the signatures of sleep disruption such as arousal patterns, frequency band alterations, and sleep stage fragmentation.

Recent advances in machine learning and deep learning have demonstrated promise for automated sleep disorder detection from physiological signals [31]. EEG signals are particularly informative for OSA detection, as apnea events trigger changes in EEG signal patterns [1], [22]. However, automated OSA classification based on EEG signals presents technical challenges. EEG signals from multichannel recordings present high-dimensional, complex spatiotemporal patterns that require sophisticated feature extraction. Furthermore, model performance must generalize across heterogeneous patient populations with varying demographics, comorbidities, and OSA severity levels to achieve clinical utility.

While several studies have applied deep learning to OSA detection using single-channel ECG and other physiological signals, fewer have comprehensively evaluated multichannel EEG approaches across diverse feature representations [28]. Recent work has explored convolutional neural networks for single-channel EEG [1], YOLO architectures for EEG spectrograms [29], and traditional machine learning with hand-crafted features [33]. However, these studies typically focus on a single feature type or architecture, limiting understanding of which signal representations best capture apnea signatures

in EEG signals. Furthermore, most prior work reports only aggregate performance metrics without stratifying results by patient demographics or sleep stages, which is an essential consideration for clinical deployment given known variations in OSA presentation across age, sex, and disease severity.

This work addresses these gaps through a comprehensive evaluation of deep learning architectures and feature representations for OSA detection from multichannel EEG. We investigate four distinct signal representations: (1) raw temporal signals, (2) short-time Fourier transform (STFT) spectrograms capturing time-frequency dynamics, (3) coherence-based graph representations encoding inter-channel functional connectivity, and (4) topological data analysis (TDA) features derived from persistent homology. TDA has shown promise in characterizing sleep-related EEG patterns [24].

We evaluate feature representations using two deep learning architectures: Vision Transformers (ViT) adapted for physiological signals, and Graph Attention Networks (GAT) for connectivity-based representations [7]. To ensure clinical relevance, we conduct rigorous stratified analysis across patient demographics (age groups, sex, AHI severity levels) and sleep stages (N1, N2, N3, REM), identifying critical factors that influence detection performance in real-world scenarios.

Our main contributions are threefold. First, we present the first comprehensive comparison of multiple deep learning architectures and feature representations for multichannel EEG-based OSA detection. Second, we introduce TDA-based features (HEPC and AP FAPC) to OSA detection and demonstrate their effectiveness in capturing apnea signatures compared to traditional representations. Third, we conduct extensive demographic and sleep stage stratification analysis, revealing performance disparities across subpopulations that have critical implications for equitable and robust clinical deployment.

II. RELATED WORK

There has been significant recent work on sleep apnea detection from EEG signals. Prior work by Mahmud et al. (2021) found that variational mode decomposition can successfully decompose EEG signals for sleep apnea classification from EEG signals alone [21]. Combined with a CNN-BiLSTM deep learning model, the authors achieved average F1 scores greater than 0.90 across three datasets of adult subjects using a subject-independent evaluation method. While this work achieved high F1 scores, the datasets used were small, with the largest dataset containing 25 mostly male subjects. Other work by Barnes et al. (2022) developed a CNN for single-channel sleep apnea detection [1]. This model achieved an accuracy of 69.9% and an average AUC across 10 folds of 0.804 on three datasets of adult subjects. Further work by Zhao et al. (2021) used traditional deep learning models on two EEG channels combined with manual feature extraction on 30 subjects [33]. While this study achieved good accuracy of 88.99%, little information about the demographics of the training and testing set was reported. Work by Tanci et al. (2025) attempted to use an object detection network, YOLOv8,

on Short-time Fourier Transform (STFT) EEG features to detect sleep apnea events, achieving an overall accuracy of 93.7% [29]. The subject demographics are not reported, but the authors stratified their model results based on sleep apnea severity, revealing consistent F1 scores across sleep apnea severity. Recent work by Irueta et al. (2025) achieved a best AUC of 0.689 for EEG-only detection of sleep apnea events using a CNN on EEG signals [13]. Other work by Manjunath et al. (2024) used Short-time Fourier transforms and CNNs to predict oxygen desaturations from 30-second EEG signals in pediatric patients, achieving a best balanced accuracy of 66.8% [22]. Critically, this study was conducted on pediatric patients rather than adults, and accounted for age and sex in the training and testing splits.

III. METHODS

A. Dataset

This work uses the Nationwide Children’s Hospital Sleep DataBank (NCHSDB) dataset with 3984 PSG studies from 3673 pediatric patients [11], [15], [16]. This dataset contains full PSG studies, including EEG, pulse oximetry, and nasal and oral sensors, among others. The dataset furthermore contains technician labels for sleep stage and sleep breathing events, including oxygen desaturations, and central, mixed, and obstructive apnea events. We use studies collected from subjects with 256 Hz EEG sampling rate and 7-channel EEG studies. In this work, we use 30-second groups of 7-channel EEG signals sampled at 256 Hz. After filtering our dataset to subjects with the appropriate 7 EEG channels and 256 Hz sampling rate, we were left with 2985 subjects. We label a particular 30-second window as “disordered breathing” (positive) or “normal breathing” (negative) if the apnea, hypopnea, oxygen desaturation label in the positive case or normal breathing label in the negative case covers 75% of the window. Data was split by subject, i.e. no data from one subject was in the training and validation sets. Furthermore, we perform an age and sex stratified train/validation split, meaning that the training set and the validation set have the same ratio of each age and sex group. Based on this labeling strategy, our training set contains 71,547 positive and 2,130,666 total labels (3.36% positive), while the test set contains 17,066 positive and 505,817 total labels (3.37% positive).

Apnea-hypopnea index (AHI) is a common metric used to classify sleep apnea severity. It is calculated by counting the total number of apnea and hypopnea events over a night of sleep, then dividing by the total sleep time. In pediatric subjects, $AHI < 1$ is considered healthy, $1 \leq AHI < 5$ is considered mild sleep apnea, $5 \leq AHI < 10$ is considered moderate sleep apnea, and $AHI \geq 10$ is considered severe sleep apnea [25]. We use these criteria to stratify subjects by sleep apnea severity in Section IV-C.

B. Feature Extraction Approaches

1) *Raw EEG Signals*: Our first feature representation are raw EEG signals. We filter powerline noise with a 60 Hz and 120 Hz 3rd order Butterworth bandstop filter. Since we are

focused on 30-second groups of 7-channel EEG data collected at 256 Hz, this feature yields a feature array $\mathcal{F} \in \mathbb{R}^{7 \times 7681}$. We apply no further filtering to the raw signals, and do not address baseline drift or apply any data deletion or interpolation procedures.

2) *Short-Time Fourier Transform (STFT)*: The second featurization technique we test are short-time Fourier transforms. We take our raw EEG signals and perform powerline noise filtration, then calculate an STFT with 129 frequency bins (0.5 - 128 Hz) using a Hann window function. Specifically, we calculate the following discrete STFT for each channel m :

$$\mathcal{S}_m[q, p] = \sum_{k=0}^{N-1} X_m[k] \overline{w[k - ph]} e^{-i2\pi qk/N} \quad (1)$$

where $q, p \in \mathbb{Z}$ denote the time and frequency index respectively and $\overline{w}$ is the complex conjugate of the Hann window function, and X_m is the discretely sampled m th channel of the signal. This process yields $\mathcal{S} \in \mathbb{C}^{129 \times 62}$, a complex-valued array with 129 frequency bins and 61 time bins. The STFT array magnitude is log-transformed and Z-normalize yielding an array $\mathcal{F} \in \mathbb{R}^{7 \times 129 \times 62}$ corresponding to 7 channels

3) *Coherence-Based Graph Construction*: To test out features which integrate functional connectivity metrics, we construct a graph representation based on our multichannel EEG signal. The graph contains 7 nodes, one for each EEG of the 7 electrodes. The signal coherence between any two channels represents the weight of the edge between the two corresponding nodes, and the STFT of that channel, calculated as in Subsection III-B2 are the node features. To calculate coherence, we take our set of time-series signals $X_i(t)$, where $i \in V = \{1, \dots, 7\}$ represents the channels and $t \in \{1, \dots, T\}$ are the time samples of our signal. We first calculate a smoothed periodogram $\hat{f}_{m,n}$ between channels m and n . We then calculate the magnitude squared signal coherence between all channels m, n as follows:

$$\mathcal{C}(X_m, X_n, \omega_k) = \frac{|\hat{f}_{m,n}(\omega_k)|^2}{\hat{f}_{m,m}(\omega_k) \hat{f}_{n,n}(\omega_k)} \quad (2)$$

This yields a feature matrix $\mathcal{C} \in \mathbb{R}^{7 \times 7 \times 129}$, representing the functional connectivity of EEG channels at each frequency bin. We use the vector $\mathcal{C}_{m,n}$ to represent the edge weight between any two channels in the 7×7 fully connected graph represented by $\mathcal{C}$, and use the STFT values $\mathcal{F}_m \in \mathbb{R}^{129 \times 62}$ calculated for each of our $m \in \{1, \dots, 7\}$ channels as in Section III-B2 as the node features for our graph representation.

4) *Topological Data Analysis Features*: Topological features were extracted using persistent homology applied to EEG coherence matrices as described in Subsection III-B3. Recall that based on our coherence matrix $\mathcal{C}$, two perfectly correlated signals have coherence 1, while two completely uncorrelated signals have coherence 0. In order to create a coherence dissimilarity matrix, we calculate a new matrix $\mathcal{D} = 1 - \mathcal{C}$, yielding a matrix where two perfectly correlated signals have dissimilarity 0, while two perfectly uncorrelated signals have

dissimilarity 1. We then apply *persistent homology* to our coherence dissimilarity matrix, using a Rips filtration to generate a persistence diagram for each frequency band. For further details, we follow the approach of [24].

Persistent homology aims to measure the topological invariants of a space, as measured through homology groups. Homology measures topological invariants such as connected components (0th homology group or H_0), holes (1st homology group or H_1), and other properties in higher dimensions. For this work, we focus on H_0 and H_1 . We first build a *filtration* of our functional connectivity, formally a Rips filtration. A filtration is a sequence of nested subsets of data. We form these subsets by varying a filtration parameter, and considering any two points with distance as defined by $\mathcal{D}$ less than the filtration parameter as connected. From the changing connectivity pattern as we increase the filtration parameter, we can calculate the topological features of the data. The scalar filtration parameter value where a particular topological feature is first observed is called the birth value, and the scalar filtration parameter value where the topological feature is no longer observed is called the death value. These (birth, death) pairs can be plotted in $\mathbb{R}^2$ to form an object called a persistence diagram which represents the topological features of the coherence dissimilarity space. The persistence diagram is a multiset of all birth and death values observed from a space. Further information on the persistent homology pipeline can be found in [10].

Persistence diagrams are poorly suited to machine learning methods, as they cannot be embedded in a Hilbert space as required by many machine learning algorithms [4]. Recent research has proposed Hermite Expansion of Persistence Curves (HEPC) and Arbitrary Period Fourier Approximation of Persistence Curves (AP-FAPC) to featurize persistence diagrams appropriately for machine learning. In order to construct these representations, we must first construct a persistence curve.

Definition III.1 (Lifespan Entropy Persistence Curve [6]). *Define the function $L : \mathbb{D} \rightarrow \mathbb{R}$, with $\mathbb{D}$ the space of persistence diagrams.*

$$L = \sum_{(b_i, d_i) \in D} (d_i - b_i) \quad (3)$$

for a persistence diagram D . We define our $\psi(b, d)$ function using Equation (3):

$$\psi(b, d) = -\frac{d - b}{L} \log \left(\frac{d - b}{L} \right) \quad (4)$$

We define the Lifespan Entropy Persistence Curve as:

$$P(D)(x) = - \sum_{(b_i, d_i) \in D} \frac{d_i - b_i}{L} \log \left(\frac{d_i - b_i}{L} \right) \chi_{[b_i, d_i)}(x) \quad (5)$$

for $x \in \mathbb{R}$.

The persistence curve is a compactly supported step function on $\mathbb{R}$. Once we have this function which can approximate a persistence diagram appropriately for machine learning applications, we are left to decide how to vectorize it. Two

popular approaches exist: Hermite Expansion of Persistence Curves (HEPC), and Arbitrary Period Fourier Approximation of Persistence Curves (AP-FAPC) [5], [23]. Briefly, these approximate the persistence curve using Hermite function coefficients in the case of HEPC, and Fourier coefficients in the case of AP-FAPC. Prior work has shown AP-FAPC to complement HEPC approximations and lead to greater accuracy when combined together, as AP-FAPC allows for finer grain approximation of the underlying persistence curve. We calculate 16 coefficients for each featurization type. In the case of HEPC, we obtain an array of size $\mathcal{F} \in \mathbb{R}^{2 \times 129 \times 16}$, for two homology groups H_0 and H_1 , 129 frequency bins, and 16 coefficients. Each AP-FAPC coefficient was decomposed into real and imaginary components, yielding 16 additional coefficients per homology dimension per frequency bin. The final feature vector concatenated HEPC coefficients with real and imaginary parts of AP-FAPC ($16 + 16 + 16 = 48$ coefficients per homology group per frequency bin), resulting in a feature array of dimensions $\mathcal{F} \in \mathbb{R}^{2 \times 129 \times 48}$ that approximates the persistence curve shape.

C. Model Architectures

1) *Vision Transformer*: As a primary architecture for all features aside from Coherence-based graph features, we use a Vision Transformer adapted for multichannel EEG analysis [8]. The model begins with instance normalization across channels, followed by patch extraction that divides the input into separate patches. For 2D inputs (STFT spectrograms, TDA features), patches of size 16×16 were extracted from inputs of shape (C, H, W) , while for 1D inputs (raw signals), patches of size 256 were extracted from inputs of shape (C, L) . Each patch was linearly projected to a 1024-dimensional embedding space and combined with learnable positional embeddings to preserve spatial information. The core architecture consisted of 24 transformer blocks, each containing layer normalization, multihead self-attention with 16 attention heads, followed by a two layer MLP. We use global average pooling to aggregate information across all patches, followed by a two layer MLP head and a final linear classifier for binary classification. For HEPC topological features with shape $(2, 129, 16)$ representing 2 homology dimensions $\times$ 129 frequency bins $\times$ 16 coefficients, the model treated the two homology groups (H_0 and H_1) as channels, extracting patches from the frequency and coefficient representation to learn relationships between topological features across frequency bands.

2) *Graph Neural Network*: In order to optimally process our coherence-based graph features, we use a Graph Attention Network (GAT) [3]. Edge features were processed through a fully connected encoder network, while node features were initially projected to 512 dimensions. The core architecture consisted of four stacked GATv2 layers with eight attention heads and hidden dimensions of 512. We use multihead attention mechanisms to weight information from neighboring channels based on both node features and coherence edge attributes. We use pooling with learnable attention to aggregate

node representations, followed by a fully connected classifier for binary classification. This architecture enabled the model to capture both local channel interactions and global brain connectivity patterns relevant to sleep apnea detection.

D. Model Training

We use the Adam optimizer to train our models [14]. In order to combat the significant class imbalance (6% apnea label vs 94% normal breathing label) we use inverse frequency class weighting on our loss function. In order to characterize the performance of our models, we used balanced accuracy (BA) and area under receiver operating characteristic curve (AUC) as evaluation metrics. These two metrics allow us to characterize the performance of our model on both the majority and minority classes, and have been extensively used in biomedical machine learning literature [30]. Hyperparameter tuning was performed through grid search. All models were trained for 10 epochs using a one-cycle learning rate [27]. All models were trained with batch size 256, except for the GAT model, which used a batch size of 1024. The models trained in STFT and EEG timeseries data used a learning rate of 0.00001, while the models trained on HEPC and HEPC + AP-FAPC features were both trained using a learning rate of 0.0001. The GAT model was trained with a learning rate of 0.000001. These hyperparameters were selected by grid search.

To ensure that the model receives appropriate training data from each demographic of subjects, we use a subject-wise stratified train/test split, with 75% train/25% test split. This ensures that an equal proportion of the training and testing data is represented by each age and sex group, i.e. the same ratio of age 5 male subjects is included in the training set and the testing set. Furthermore, the data split is conditioned on subjects, meaning that all data from each subject is in either the train or test sets, but not both. For this manuscript, we evaluate on a single train/test split; K-fold cross validation is left for later work.

IV. RESULTS AND DISCUSSION

A. Overall Comparative Performance Analysis

Table I presents overall test set performance. We report area under ROC curve (AUC), balanced accuracy (Bal. Acc.), sensitivity (Sens), and specificity (Spec), as well as the number of parameters in the model (# Params). Topological features (HEPC) achieved the highest performance (AUC 0.750), narrowly outperforming the GAT model trained on coherence representations and exceeding traditional time-frequency features. The GAT demonstrated the highest sensitivity and balanced accuracy despite using 96.8% fewer parameters than transformer-based approaches. This demonstrates that explicit encoding of EEG channel relationships as graph structure is substantially more parameter-efficient than learning these relationships implicitly through transformer self-attention. Notably, augmenting HEPC with AP-FAPC coefficients degraded performance (AUC 0.740), suggesting the additional topological complexity introduces noise rather than complementary information. The raw EEG baseline continued to show poor

TABLE I
OVERALL PERFORMANCE COMPARISON ACROSS METHODS AND FEATURE REPRESENTATIONS

Model	Feature	# Params	AUC	Bal. Acc	Sens	Spec
Transformer	EEG Timeseries	308,376,577	0.639	0.606	0.609	0.603
Transformer	STFT	308,370,433	0.736	0.670	0.752	0.588
GAT	Coherence Graph	9,979,129	0.748	0.688	0.767	0.610
Transformer	HEPC	307,043,329	0.750	0.685	0.732	0.639
Transformer	HEPC + AP-FAPC	307,059,713	0.740	0.674	0.728	0.619

performance (AUC 0.639), confirming the necessity of structured feature representations for this task.

The superior performance of HEPC features and the GAT model suggests that graph representations of EEG coherence patterns capture sleep apnea signatures more effectively than direct spectral analysis. Recall that the HEPC features capture the topological features of our coherence graph, without including the extra STFT node features passed to the GAT. Persistent homology extracts multiscale topological features that may better characterize the disrupted functional connectivity patterns associated with apnea events and cortical arousals. The Hermite expansion provides a parameterization of persistence curves that appears well suited to the coherence dissimilarity matrices computed from multichannel EEG. While a best-performance AUC of 0.750 is not sufficient for clinical application, this study represents one of the first analyses of multiple features across similar architectures for the task of sleep apnea identification from EEG. Further work is required to improve classification performance.

B. Performance Across Patient Demographics

Figure 1 presents age and sex stratified model performance. Female subjects demonstrated consistently superior performance across all successful models. This sex-based disparity is present in all model types. The raw EEG baseline exhibited a gap between the male and female groups alongside poorest overall performance, indicating that the sex bias is not an artifact of specific feature representations but rather reflects underlying differences in either OSA presentation or EEG signal characteristics between male and female subjects. This result is even more surprising given that we had significant numbers of both male and female subjects (330 male, 245 female), similar AHI levels in each group (average AHI 2.211 for males and 2.117 for females), and no statistically significant difference in age distributions ($p = 0.416$ for a Mann-Whitney U-test of male and female age distribution in the test set).

Performance on individual age groups peaked at age 8 and 10, while age groups 2-6 and 13-18 showed notably degraded performance. All models exhibited performance fluctuations across age groups. The raw EEG model showed the most extreme fluctuation. These findings underscore the importance of demographic aware model validation, as aggregate performance metrics obscure substantial heterogeneity that could impact clinical deployment across diverse patient populations.

TABLE II
MODEL PERFORMANCE STRATIFIED BY OSA SEVERITY

Feature	Severity	n	AUC	Bal. Acc	Sens	Spec
EEG Timeseries	Healthy	383	0.650	0.615	0.629	0.602
	Mild	133	0.651	0.615	0.616	0.614
	Moderate	33	0.592	0.564	0.543	0.585
	Severe	26	0.590	0.565	0.558	0.571
STFT	Healthy	383	0.748	0.680	0.758	0.601
	Mild	133	0.731	0.671	0.780	0.562
	Moderate	33	0.664	0.605	0.709	0.500
	Severe	26	0.706	0.650	0.694	0.606
Coherence Graph	Healthy	383	0.759	0.704	0.781	0.626
	Mild	133	0.754	0.697	0.819	0.575
	Moderate	33	0.670	0.623	0.722	0.524
	Severe	26	0.697	0.623	0.597	0.649
HEPC	Healthy	383	0.754	0.691	0.725	0.657
	Mild	133	0.743	0.681	0.749	0.614
	Moderate	33	0.706	0.645	0.780	0.510
	Severe	26	0.707	0.641	0.660	0.623
HEPC + AP-FAPC	Healthy	383	0.744	0.679	0.723	0.635
	Mild	133	0.743	0.682	0.759	0.605
	Moderate	33	0.689	0.613	0.750	0.475
	Severe	26	0.690	0.625	0.641	0.609

C. Performance Across Sleep Apnea Severity

OSA severity stratification presented in Table II revealed distinct performance patterns across the disease spectrum. Bold values indicate best performance for an AHI severity level. Across STFT, coherence graph, HEPC, and HEPC+AP-FAPC models, performance was highest on healthy subjects and progressively degraded with mild and moderate OSA. However, for the STFT and coherence graph models, the severe OSA increased significantly from the healthy and mild groups. For the HEPC and HEPC+AP-FAPC models, the model performance on the severe AHI remained similar to the performance in the moderate AHI level.

D. Performance Across Sleep Stages

Sleep stage stratification presented in Table III revealed performance variations across sleep stages. Bold values indicate best performance across sleep stages. N2 and N3 sleep stages demonstrated the highest overall performance, with HEPC achieving AUC in N2, and HEPC+AP-FAPC achieving highest AUC in N3 sleep. Wake and N1 light sleep showed poorest aggregate performance. Notably, wake and N1 epochs produced high sensitivity but poor specificity, indicating systematic over prediction of disordered breathing during arousal-prone states. Conversely, N3 deep sleep yielded lower sensitivity but high specificity, suggesting models miss apnea events that fail to

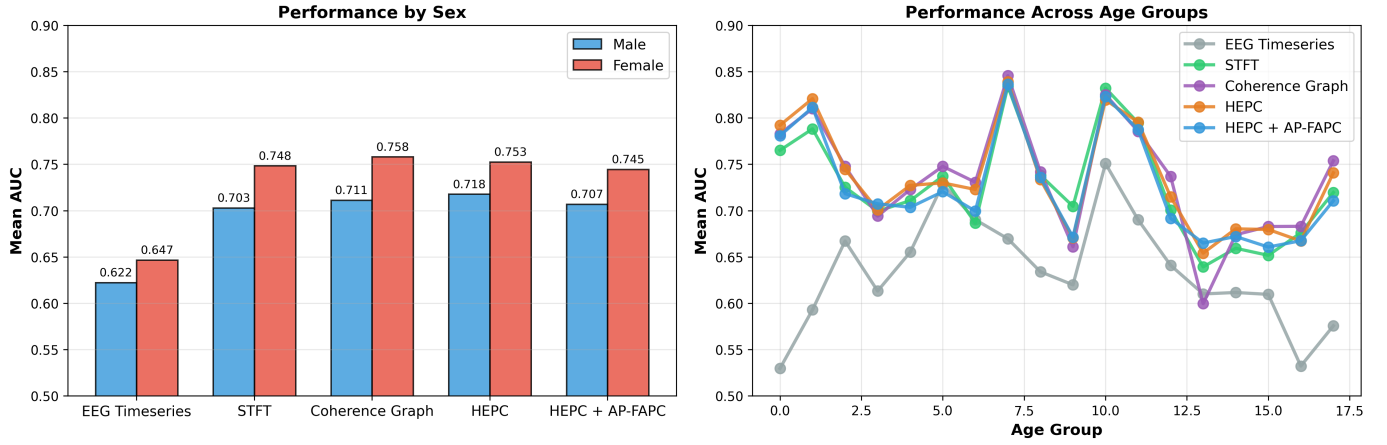


Fig. 1. Model performance across age and sex groups

TABLE III
MODEL PERFORMANCE STRATIFIED BY SLEEP STAGE

Feature	Stage	AUC	Bal. Acc	Sens	Spec
EEG Timeseries	Wake	0.545	0.539	0.619	0.458
	N1	0.568	0.552	0.710	0.395
	N2	0.594	0.570	0.571	0.568
	N3	0.539	0.505	0.090	0.919
	REM	0.596	0.574	0.779	0.369
STFT	Wake	0.688	0.644	0.857	0.431
	N1	0.663	0.633	0.803	0.462
	N2	0.704	0.636	0.684	0.589
	N3	0.705	0.614	0.491	0.737
	REM	0.705	0.654	0.766	0.541
Coherence Graph	Wake	0.702	0.649	0.900	0.398
	N1	0.664	0.642	0.845	0.439
	N2	0.697	0.641	0.718	0.564
	N3	0.697	0.564	0.242	0.886
	REM	0.712	0.668	0.798	0.538
HEPC	Wake	0.699	0.648	0.868	0.428
	N1	0.678	0.642	0.789	0.495
	N2	0.719	0.646	0.623	0.669
	N3	0.732	0.637	0.478	0.795
	REM	0.721	0.666	0.754	0.578
HEPC + AP-FAPC	Wake	0.693	0.645	0.857	0.434
	N1	0.670	0.635	0.781	0.489
	N2	0.708	0.634	0.629	0.638
	N3	0.734	0.647	0.538	0.757
	REM	0.715	0.655	0.726	0.583

trigger cortical arousals. This sensitivity and specificity profile across the sleep stages demonstrates that EEG-based detection fundamentally captures arousal-associated apnea rather than pure respiratory obstruction.

The sleep stage stratification results further suggest that our models primarily detect EEG signatures associated with arousal-triggering apnea events rather than respiratory obstruction. Across all successful models, we observe an inverse relationship between sensitivity and specificity: Wake and N1 stages exhibit high sensitivity (> 0.80 for STFT, coherence graph, and HEPC) but poor specificity (< 0.50), while N3 stage shows the opposite pattern with sensitivity dropping to 0.478 (HEPC) or lower, while specificity rises to 0.795-0.886. This trade-off aligns with known physiology of arousal

thresholds across sleep stages. In light sleep, cortical arousal is easily triggered, the model flags many true positive events but also misclassifies non-apnea arousals as disordered breathing [9]. Conversely, in N3 sleep, where the arousal threshold is physiologically higher and many apnea events fail to produce a cortical response, the model misses most events yet maintains high specificity when it does detect something [9]. Our positive labels are defined by technician-annotated apnea/hypopnea and oxygen desaturation events rather than isolated arousals. The classifiers appear to have learned a combination of cortical activation patterns and disrupted connectivity that accompanies respiratory events severe enough to trigger arousal. This interpretation is supported by the superior performance of TDA features (HEPC), which capture global network topology of coherence rather than local arousal transients, suggesting the model exploits sustained, multiscale changes in functional connectivity induced by intermittent hypoxia and sleep fragmentation.

V. CONCLUSION

In this work, we present a comprehensive evaluation of deep learning architectures and feature representations for automated sleep apnea detection from multichannel EEG, with rigorous stratification across patient demographics, OSA severity levels, and sleep stages. Topological data analysis features (HEPC) achieved the highest overall performance, outperforming traditional time-frequency representations and demonstrating that persistent homology effectively captures central nervous system manifestations of sleep-disordered breathing. However, demographic stratification revealed critical performance heterogeneity: models consistently performed better on female subjects, model performance on moderate OSA was unexpectedly poor, and sleep stage analysis uncovered high sensitivity and low specificity in lighter sleep and low sensitivity and high specificity in deeper sleep. These findings demonstrate that EEG-based detection fundamentally capture arousal-associated apnea rather than respiratory obstruction, with performance strongly dependent on sleep

architecture and patient characteristics. While our results advance automated OSA detection methodology and establish topological features as a promising approach, the substantial performance variability across demographic and physiological subgroups underscores the need for demographic aware model development and larger multicenter validation before clinical deployment. Future work should validate these approaches on external datasets and explore ensemble methods leveraging complementary features across patient populations.

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